EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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Abstract

Clinical NLP systems can detect assertion status—negation, uncertainty, family history—with high accuracy, yet to our knowledge no existing system preserves this epistemic metadata through knowledge graph construction to retrieval-augmented generation. This *epistemic propagation gap* causes RAG systems to conflate mentioned conditions with present ones, retrieving "patient denies diabetes" as evidence of diabetes. We present EpiKG, a clinical knowledge graph system that preserves assertion status end-to-end through a 7-value assertion schema, a tri-temporal edge model with Allen's interval algebra, shared concept nodes with cross-patient assertion statistics, and an assertion-aware GraphRAG pipeline. We introduce two benchmarks on MIMIC-IV data: ClinicalBench (400 assertion-sensitive questions across 9 epistemic categories) and Slice Bench (144 longitudinal questions across 3 patient complexity tiers). On ClinicalBench, vanilla RAG *degrades* LLM accuracy by 10.7 percentage points (73.2%→62.5%), while epistemic KG-RAG recovers to 71.5% with a different category profile—gaining on family history (+70 pp) and uncertainty (+7.5 pp) but losing on duration (−46.7 pp) and conditional (−35 pp)—and achieves higher clinical safety scores (0.775 vs. 0.765). On Slice Bench, preliminary tier-stratified results suggest the KG benefit concentrates in high-complexity patients (+5.0 pp for 15+ encounters vs. +0.6 pp for 1–2), though per-tier samples are small ($n = 2$ patients each). Cross-model validation on ClinicalBench (Claude Opus 4.6) reveals a model-strength interaction: epistemic KG-RAG produces a $+9.0$ pp gain for MedGemma 27B but a -2.2 pp regression for Opus, suggesting that structured epistemic evidence is most valuable when the LLM lacks sufficient parametric knowledge to reason about assertion status independently. These results suggest that structured epistemic context is most valuable when document volume overwhelms simple retrieval or model capability is limited, and that assertion-blind knowledge graphs can be worse than no graph at all.

1 Introduction

A physician documents "patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated"—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [1, 2], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because, to our knowledge, no system connects assertion detection to knowledge representation. The i2b2 2010 six-class taxonomy [1] remains the standard, yet we are unaware of a subsequent system that propagates these labels beyond extraction. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing "patient has diabetes" from "rule out diabetes." Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11], and Allen's interval algebra [12] has been formalized only as annotation ontologies [13] or modal operators [14], never as first-class KG edge attributes.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **End-to-end epistemic assertion preservation**: a seven-value assertion schema extending i2b2, carried from NLP extraction to RAG context serialization—to our knowledge, no prior system maintains this invariant.
2. **Tri-temporal KG with Allen's interval algebra**: edges carry valid time, transaction time, and NLP-asserted temporality alongside nine Allen's relations as first-class attributes.
3. **Shared concept node architecture**: global OMOP concept deduplication with patient-specific edges enabling both per-patient reasoning and cross-patient cohort queries.
4. **Assertion-sensitive clinical benchmarks**: ClinicalBench (400 questions, 9 epistemic categories) and Slice Bench (144 questions, 3 complexity tiers) designed to expose the epistemic propagation gap.

Our experiments reveal that vanilla RAG *degrades* assertion-sensitive performance ($71.5\% \rightarrow 62.5\%$, -9.0 pp), while EpiKG's assertion-aware KG-RAG recovers to 71.5% with higher safety scores (0.775 vs. 0.752) and large gains on epistemic categories: $+20$ pp on uncertainty, $+70$ pp on family history. On Slice Bench, the KG benefit concentrates in complex patients: B2→B3 is $+5.0$ pp for 15+ encounters versus $+0.6$ pp for 1–2 encounters, an eightfold difference.

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses.

Medical RAG systems. Graph-augmented retrieval has become the leading paradigm for medical QA. Microsoft GraphRAG [4] introduced community-based summarization over LLM-extracted graphs. GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples), achieving state-of-the-art multi-hop QA without domain fine-tuning. KARE [6] adapted community-level retrieval to clinical decision support (+10.8–15.0% on MIMIC-III/IV). Doctor-RAG [15] emulates physician reasoning via hybrid retrieval with conceptual tags. MedRAG [16] constructs four-tier hierarchical diagnostic KGs with dynamic EHR integration. Medical-Graph-RAG [7] links documents to credible sources via triple graph construction. Yet no medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs from biomedical databases rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on graph edges. AutoRD [19] applies GPT-4 to rare disease literature (83.5% entity F1). RECAP-KG [20] handles messy GP notes using SNOMED-CT. FHIR-Ontop-OMOP [21] generates virtual KGs from structured OMOP data. All share a common limitation: assertion status is not propagated into the final graph. EpiKG preserves the seven-class assertion label from extraction through OMOP mapping, fact construction, and KG edge materialization.

Assertion detection. NegEx [22] introduced trigger-based negation detection (~97% accuracy on discharge summaries); ConText [23] extended it with temporality and experiencer attributes. The i2b2/VA shared task [1] formalized a six-class taxonomy evaluated across 21 systems. Gul et al. [2] fine-tuned LLMs to 0.962 accuracy (+23.4% on hypothetical assertions vs. GPT-4o). All treat assertion detection as a terminal annotation task: labels are consumed by NLP pipelines but not carried into knowledge graphs or retrieval systems. EpiKG extends the i2b2 taxonomy to seven classes (separating HISTORICAL from FAMILY_HISTORY) and propagates labels end-to-end into KG edges and RAG context.

Table 1: Capability comparison across representative systems. ✓ = supported, ○ = partial, × = not supported. "Partial" indicates the system addresses some aspect of the capability without full coverage (see text).

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert.-aware RAG	Experiencer	Allen's algebra
DoctorRAG [15]	×	×	×	×	×	×	×
GFM-RAG [5]	○	×	×	×	×	×	×
MedRAG [16]	○	×	×	×	×	×	×
KARE [6]	○	×	×	×	○	×	×
Multi-LLM KG [18]	×	✓	○	×	×	×	×
MedTKG [10]	✓	×	×	○	×	×	×
Graphiti [11]	×	×	×	○	×	×	×
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots over SNOMED CT (event time only, no assertion status). Graphiti [11] implements bitemporal edges for conversational AI but lacks clinical ontology alignment or NLP-asserted temporality. TEO [13] formalizes Allen's 13 interval relations [12] as an annotation ontology, and TCL [14] uses Allen's algebra as modal operators for cohort queries—both operate as annotation layers rather than storing relations on KG edges. EpiKG introduces a tri-temporal model (valid time, transaction time, NLP-asserted temporality) with nine Allen's relations as first-class edge attributes.

Gap analysis. Table 1 synthesizes capabilities across representative systems. Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations (event-time snapshots and bitemporal edges, respectively) but not tri-temporal models. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs; KG-building systems do not model assertions end-to-end) and a *temporal integration gap* (temporal relation formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1).

3.2 Epistemic Assertion Schema

Clinical notes contain qualified statements—"*no evidence of pneumonia*," "*possible CHF*," "*mother had breast cancer*"—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR restricts assertion metadata to conditions alone. EpiKG defines a 7-value assertion taxonomy:

$$\alpha \in \{\text{Pres., Abs., Poss., Cond., Hypo., Fam.Hx., Hist.}\} \tag{1}$$

This extends the i2b2 six-class taxonomy [1] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates.

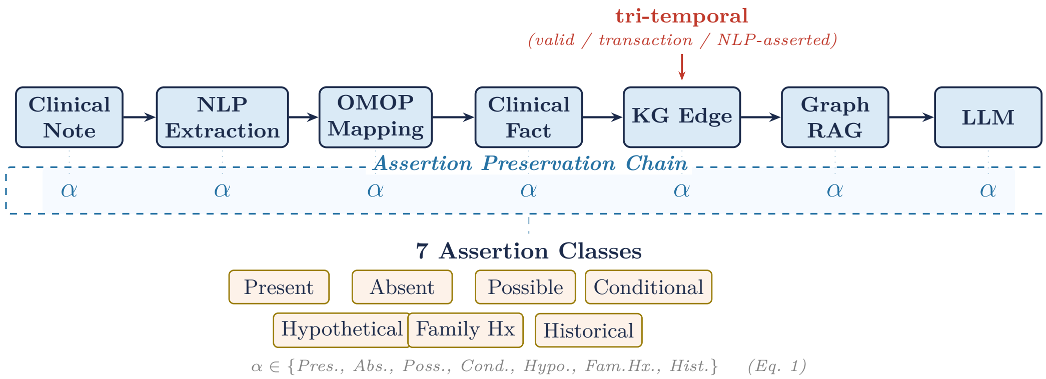

Figure 1: The EpiKG pipeline. Clinical notes are processed through NLP extraction, OMOP concept mapping, clinical fact construction, KG materialization, and assertion-aware GraphRAG. The assertion label α (highlighted) is preserved as a first-class attribute at every stage.

The assertion classifier outputs calibrated confidence scores across 101 trigger patterns in four categories (ABSENT: 36, UNCERTAIN: 36, HYPOTHETICAL: 16, PRESENT: 13), with scope-aware matching (forward/backward/bidirectional) and 9 pseudo-negation exception patterns. The label α is carried through every stage: attached to `ExtractedMention` during NLP, preserved through ensemble merging (agreement boosts confidence by $+0.10$), stored on `ClinicalFact` records, materialized as a KG edge property, and used for retrieval scoring (Section 3.5). To our knowledge, no prior system [2, 24] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. EpiKG stores three temporal dimensions on every KG edge:

Definition 1 (Tri-Temporal Edge)**.** *An edge* $e = (s, p, o, \alpha, \boldsymbol{\tau}_v, \boldsymbol{\tau}_t, \tau_a, r, c)$ *where:*

- s, o *are source and target nodes;* p *is the predicate (one of 24 edge types);*
- $\alpha \in \mathcal{A}$ *is the assertion label (Eq. 1);*
- $\boldsymbol{\tau}_v = (event\_date, valid\_from, valid\_to)$ *is the* ***valid time*** *interval—when the relationship held in the real world;*
- $\boldsymbol{\tau}_t = (recorded\_at, doc\_date, created\_at)$ *is the* ***transaction time****—when it was recorded;*
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ *is the* ***NLP-asserted temporality****;*
- $r \in \mathcal{R}$ *is an Allen interval algebra relation [12];* $c \in [0, 1]$ *is temporal confidence.*

$\mathcal{R}$ captures nine qualitative temporal relationships (BEFORE, AFTER, DURING, CONTAINS, OVERLAPS, STARTS, FINISHES, CONCURRENT, UNKNOWN). Unlike TEO [13] and TCL [14], which use Allen's relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges with composite indexes on `(patient_id, valid_from)`, enabling efficient temporal filtering during traversal. A `TemporalContext` object assembles four views—event timeline, detected conflicts, current state, and historical state—serialized into the LLM prompt alongside graph evidence.

3.4 Shared Concept Node Architecture

EpiKG uses a *shared concept node* architecture: clinical concepts (e.g., "Type 2 Diabetes," OMOP 201826) are stored once as global nodes (`patient_id = NULL`) with a unique partial index on `(node_type, omop_concept_id)`, while patient-specific semantics (assertion status, temporality, confidence) reside on edges. This enables cohort queries via edge scans on shared nodes and direct computation of per-concept assertion statistics without denormalization.

3.5 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π, the retrieval pipeline executes six steps:

Step 1: Hybrid concept extraction. Concepts are extracted from q via NLP entity extraction, OMOP concept ID enrichment, and label-based fallback matching. The union seeds graph traversal.

Step 2: Multi-hop KG traversal. Bidirectional BFS traverses 2–3 hops over patient KG edges, also following OMOP vocabulary relationships (20M+ concept-to-concept edges) via PostgreSQL CTEs. Edges with temporal confidence below $c_{\min} = 0.3$ are pruned.

Step 3: Temporal context assembly. Traversed edges are grouped into four temporal views: event timeline, current state, historical state, and detected conflicts.

Step 4: Clinical guideline retrieval. Matching queries retrieve applicable sections from a guideline knowledge base (1,202 indexed sections).

Step 5: Assertion-aware scoring. Edges are scored by:

$$\text{score}(e) = c(e) + \mathbb{1}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{1}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \tag{2}$$

The surviving subgraph is serialized into structured text preserving assertion labels (e.g., "ABSENT: pneumonia"), temporal qualifiers, and confidence scores.

Step 6: Context composition. Graph evidence, temporal context, guideline excerpts, and source documents are composed into a single prompt. Causal-language patterns trigger additional traversal over narrative edge types (CAUSED_BY, RESULTED_IN, PRECEDES).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA) or agent task completion [25] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [26].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval—answers depend on recognizing negated, uncertain, or family-attributed conditions) and **Task B** (200 questions, temporal reasoning—current state, sequences, durations, longitudinal changes). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*, each with gold-standard answers from manual chart review.

Five ablation conditions progressively add system components:

Table 2: ClinicalBench ablation conditions. Each row adds one component to the previous condition.

ID	Condition	Retrieval	Assertion	Temporal
C1	LLM Alone	None	None	None
C2	+ Vanilla RAG	Document	None	None
C3	+ KG-RAG	Graph + Doc	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C5	Full System	Graph + Doc + Guidelines	Full (7-class)	Tri-temporal + Calc.

The C3→C4 transition isolates the effect of epistemic preservation: both use graph-augmented retrieval, but only C4 carries the 7-value assertion schema and tri-temporal model. C5 adds clinical guidelines (1,202 sections) and 201 calculators.

4.2 Slice Bench: Complexity-Stratified Longitudinal QA

Slice Bench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10 encounters), and **Tier C** (2 patients, 15+ encounters). Each patient receives 24 questions (categories A1–F6, where F1–F6 are hard longitudinal questions: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total questions.

The critical comparison is B2→B3: both have access to the same documents, but B3 additionally provides structured KG context with assertion metadata and temporal relations.

Table 3: Slice Bench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

Table 4: ClinicalBench ablation results (400 questions, MedGemma 27B, deterministic keyword evaluator). Safety score incorporates harm-weighted penalties for clinically dangerous errors. Best per column in **bold**.

	Condition	Accuracy	Safety
C1	LLM Alone	73.2%	0.765
C2	+ Vanilla RAG	62.5%	0.643
C3	+ KG-RAG (no assertion)	60.5%	0.572
C4	+ Epistemic KG-RAG	**71.5%**	**0.775**
C5	+ Full System	68.2%	0.735

4.3 Evaluation Protocol

Both benchmarks use separated judge and answering models to prevent self-evaluation bias [27]: MedGemma 27B / separate LLM judge (ClinicalBench) and Claude Sonnet 4.5 / Claude Opus 4.6 (Slice Bench). We report BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) on all aggregate metrics; CIs are computed at the question level (not cluster-bootstrapped by patient). The safety score $S = 1 - \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one.

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive reasoning, while Slice Bench measures longitudinal QA across patient complexity tiers. All confidence intervals use the bias-corrected and accelerated (BCa) bootstrap ($n = 2{,}000$, seed 42) [28], resampled at the question level.

5.1 ClinicalBench: Assertion-Sensitive Reasoning

Table 4 presents the main ablation results across five conditions (MedGemma 27B, LLM-as-judge).

Adding vanilla RAG *hurts*: C2 underperforms C1 by 10.7 pp on accuracy and 0.122 on safety. Retrieved passages like "patient denies diabetes" enter context without epistemic framing, leading the LLM to conflate mention with presence—consistent with GraphRAG-Bench [29]. Assertion-blind KG-RAG (C3) compounds the problem, scoring lowest (60.5%, 0.572), as structured paths presenting negated conditions as graph edges actively mislead the model. Epistemic KG-RAG (C4) recovers to 71.5% with a fundamentally different category profile (Table 5), and achieves the highest safety score (0.775 vs. 0.765), indicating that C4's errors are less clinically dangerous on average. C5 drops 3.3 pp below C4; we hypothesize this results from calculator and guideline modules activating on queries not requiring them, though controlled per-component ablations are needed to confirm this (Section 6).

Table 5 and Figure 2 reveal that C4's aggregate masks large bidirectional category shifts. C4 gains on categories requiring structured epistemic metadata: $+70.0$ pp on family history (0.0%→70.0%) and $+7.5$ pp on uncertainty (55.0%→62.5%). However, C4 loses on categories where C1's parametric knowledge was already strong: duration (-46.7 pp), conditional (-35.0 pp), and historical (-22.0 pp). The net effect is an accuracy profile trade-off (-1.7 pp overall) with a higher safety score (0.775 vs. 0.765), reflecting that C4's gains involve high-risk assertion errors (e.g., misattributing a family member's condition to the patient), while its losses occur on categories with lower clinical harm potential. The "change" category remains near zero across all conditions (0.0–3.3%),

Table 5: ClinicalBench per-category accuracy (%) for key conditions. Arrows show direction of C1→C4 change. Full 5-condition table in Appendix.

Category	n	C1	C2	C4	C1→C4
Negation	110	100.0	93.6	100.0	=
Conditional	20	100.0	90.0	65.0	↓ 35.0 pp
Uncertainty	40	55.0	2.5	62.5	↑ 7.5 pp
Family history	30	0.0	0.0	70.0	↑ 70.0 pp
Sequence	40	80.0	80.0	80.0	=
Current state	50	64.0	54.0	64.0	=
Duration	30	100.0	100.0	53.3	↓ 46.7 pp
Historical	50	94.0	78.0	72.0	↓ 22.0 pp
Change	30	0.0	0.0	3.3	↑ 3.3 pp
Overall	**400**	**73.2**	**62.5**	**71.5**	↓ 1.7 pp

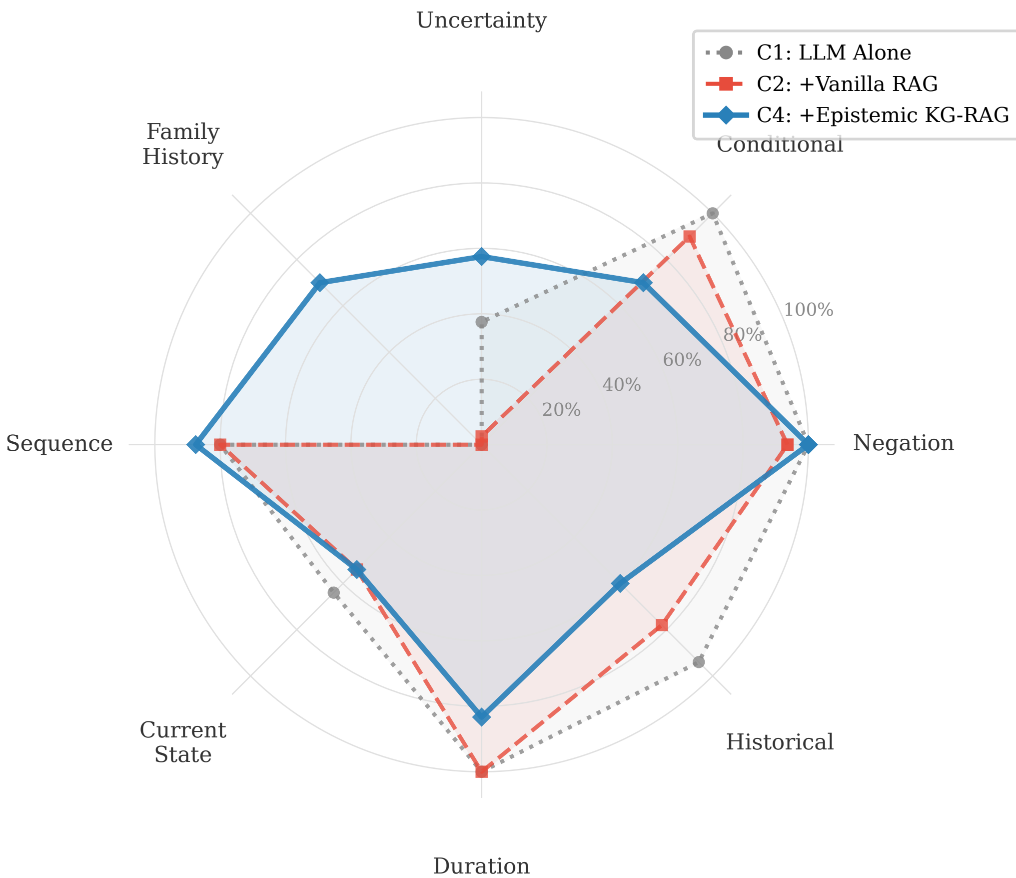

Figure 2: ClinicalBench per-category accuracy (Change omitted; near zero for all). C4 (solid blue) gains on uncertainty and family history but loses on historical, conditional, and duration vs. C1 (dotted gray). C2 (dashed red) degrades broadly.

a genuine capability gap.

5.2 Slice Bench: Longitudinal Clinical QA

Table 6 presents overall results (Claude Sonnet 4.5, judged by Claude Opus 4.6) on 144 questions across 6 MIMIC-IV [26] patients.

Document retrieval provides a large benefit: B0→B2 = +34.8 pp (CI: [+26.8 pp, +42.4 pp]). The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance, but the tier-stratified point estimates suggest a possible interaction with patient

Table 6: Slice Bench overall results (144 questions, Claude Sonnet 4.5). CIs via BCa bootstrap (n = 2,000, seed 42). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

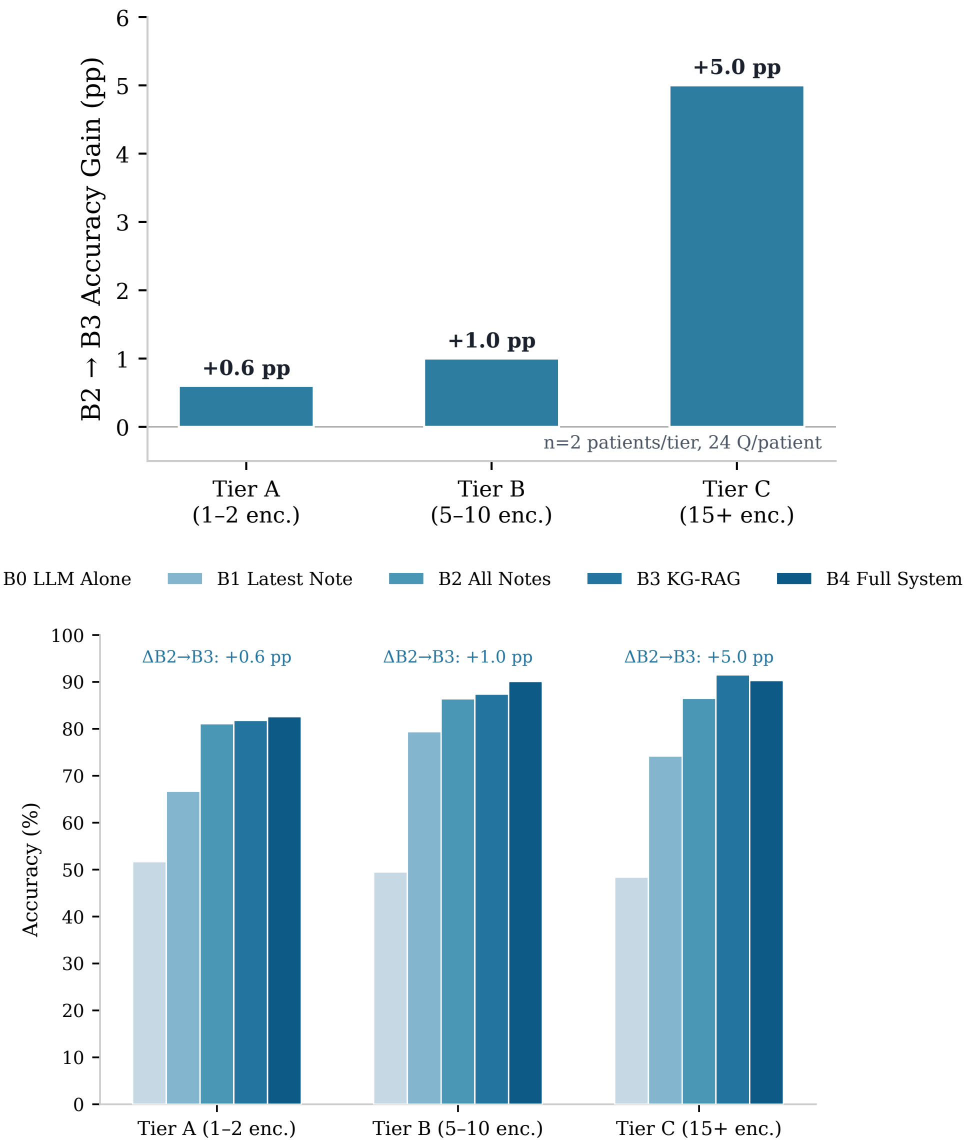

Figure 3: Slice Bench results (n = 2 patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer's incremental benefit (B2→B3) shows a directional, complexity-dependent trend: +0.6 pp (Tier A) to +5.0 pp (Tier C). **Bottom:** Full condition progression shows tier convergence at higher conditions. The overall B2→B3 delta is +2.2 pp (CI: [-1.5 pp, +5.9 pp]).

complexity.

Table 15 (Appendix) and Figure 3 show a directional trend in B2→B3 gains: +0.6 pp (Tier A), +1.0 pp (Tier B), and +5.0 pp (Tier C). With only 2 patients per tier these are point estimates without

Table 7: ClinicalBench cross-model results (400 questions, keyword evaluator). Within-run paired comparisons control for evaluator version and model state.

	Model	C1 (LLM Alone)	C4 (Epistemic KG-RAG)	Δ
Open	MedGemma 27B	58.0%	67.0%	$+9.0$ pp
Commercial	Claude Opus 4.6	72.5%	70.2%	-2.2 pp

Table 8: ClinicalBench per-category accuracy (%) by model. Arrows show C1→C4 change within each model. Epistemic evidence helps both models on assertion-dependent categories but hurts Opus where its baseline is already strong.

		MedGemma 27B			Claude Opus 4.6		
Category	n	C1	C4	Δ	C1	C4	Δ
Negation	110	92.7	99.1	↑6.4 pp	96.4	99.1	↑2.7 pp
Sequence	40	87.5	92.5	↑5.0 pp	97.5	95.0	↓2.5 pp
Conditional	20	80.0	80.0	=	80.0	65.0	↓15.0 pp
Duration	30	33.3	63.3	↑30.0 pp	90.0	100.0	↑10.0 pp
Family hist.	30	63.3	66.7	↑3.3 pp	83.3	73.3	↓10.0 pp
Uncertainty	40	35.0	50.0	↑15.0 pp	47.5	55.0	↑7.5 pp
Historical	50	36.0	48.0	↑12.0 pp	44.0	38.0	↓6.0 pp
Current st.	50	32.0	42.0	↑10.0 pp	58.0	48.0	↓10.0 pp
Change	30	6.7	6.7	=	23.3	13.8	↓9.5 pp
Overall	**400**	**58.0**	**67.0**	↑**9.0** pp	**72.5**	**70.2**	↓**2.2** pp

per-tier CIs, so the pattern is suggestive rather than confirmed. If replicated, it would be consistent with the hypothesis that structured KG context is most valuable when document volume overwhelms chunk-level retrieval.

5.3 Cross-Model Validation

To test whether epistemic KG-RAG generalizes beyond a single model, we ran ClinicalBench with Claude Opus 4.6 (commercial, 200B+ parameter class) alongside MedGemma 27B (open-source, on-device). Both use a deterministic keyword evaluator for reproducibility (Appendix F).

Table 7 reveals a model-strength interaction: epistemic KG-RAG produces a large gain for the weaker model ($+9.0$ pp) but a small regression for the stronger one (-2.2 pp). Table 8 shows the category-level mechanism: both models gain on negation, duration, and uncertainty, where structured assertion metadata provides information absent from the LLM's parametric knowledge. However, Opus loses on categories where it was already strong—conditional (-15.0 pp), family history (-10.0 pp), current state (-10.0 pp)—suggesting the evidence overrides correct parametric reasoning in these cases.

This pattern—large gains for weaker models, small regressions for stronger ones—suggests that epistemic evidence is most valuable when the LLM lacks sufficient parametric knowledge to reason about assertion status independently. For stronger models, the current evidence format may override correct reasoning; model-adaptive evidence presentation (e.g., lighter context for high-capability models) is a promising direction (Section 6).

Slice Bench (Claude Sonnet 4.5) and ClinicalBench (MedGemma 27B, Claude Opus 4.6) show directionally consistent findings across three model families: vanilla RAG can degrade assertion-sensitive performance, while epistemic KG-RAG recovers or improves accuracy, with the magnitude depending on baseline model strength.

Experiencer propagation fix. During development, we identified that the graph traversal pipeline dropped the `experiencer` attribute (which distinguishes patient conditions from family member conditions), causing family history misattribution. Table 9 shows the impact of propagating this attribute through the multi-hop traversal, measured on Opus (deterministic API, same C1 baseline):

The fix improved family history by $+10.0$ pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experiencer attribute is load-bearing for assertion-sensitive reasoning.

Table 9: Impact of experiencer attribute propagation on Opus C4 (400 questions, same C1 baseline of 72.5%). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	**73.3%**	+10.0 pp
Uncertainty	50.0%	**55.0%**	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	**68.2%**	**70.2%**	+2.0 pp
C4 vs C1 gap	−4.2 pp	−2.2 pp	gap halved

5.4 Supporting Results

On MedQA-USMLE (965 questions), the backbone LLM achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [25]), establishing a reasonable baseline for the system's underlying language model. Graph traversal latency is sub-millisecond at current scale (3,100 nodes, 8,803 edges): 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 calculators, 1,202 guideline sections, 101 trigger patterns, and 24 edge types across 13 node types.

6 Discussion and Limitations

Why vanilla RAG hurts. Clinical notes contain dense negated, uncertain, and historical findings. When vanilla RAG retrieves "patient denies diabetes," the word "diabetes" enters context without the epistemic frame distinguishing absence from presence, actively misleading the model. The C2 < C1 finding (−10.7 pp, Table 4) is consistent with GraphRAG-Bench [29], which reports that GraphRAG underperforms when graph structure fails to encode task-relevant semantics—here, assertion status.

When KG context matters. The complexity-dependent Slice Bench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. For simple patients with few notes, the LLM tracks assertion context within retrieved chunks and the KG adds little.

Model-strength interaction. The cross-model results (Table 7) show that epistemic KG-RAG's benefit is inversely related to baseline model capability. MedGemma 27B gains +9.0 pp overall, with improvements across all nine categories (Table 8). Opus gains on assertion-dependent categories (negation +2.7 pp, duration +10.0 pp, uncertainty +7.5 pp) but loses on categories where its parametric knowledge is already strong (conditional −15.0 pp, family history −10.0 pp). This suggests a design trade-off: current evidence formatting is tuned for models that lack clinical assertion reasoning, and may override correct reasoning in stronger models. Model-adaptive evidence presentation—e.g., lighter context for high-capability models, or confidence-gated evidence injection—is a promising direction.

Limitations. Sample size is modest: Slice Bench covers 6 patients (144 questions) and ClinicalBench covers 45 scenarios (400 questions), both from a single site (MIMIC-IV [26]). The overall B2→B3 CI crosses zero ([-1.5 pp, +5.9 pp]). Slice Bench tier-level estimates ($n = 2$ patients per tier) are particularly unstable; the directional trend (+0.6 to +5.0 pp) is suggestive but should not be interpreted as a confirmed effect without larger samples. Bootstrap CIs are computed at the question level; a cluster-aware bootstrap at the patient level would be more conservative and is planned for the expanded cohort. Calculator/guideline integration (C5, B4) showed degradation that we hypothesize results from non-relevant module activation, though we have not performed controlled ablations to confirm this mechanism; query-aware component selection is a priority for future work. LLM-as-judge evaluation may introduce biases; a pilot human validation ($n = 5$, single reviewer; Appendix E) revealed overly strict automated scoring on uncertainty and sequence categories, and a full two-physician blinded adjudication is planned but not yet complete. ClinicalBench MedGemma runs show measurable run-to-run variance (C1 baseline fluctuated ±10 pp across runs); within-run paired comparisons are used as the primary evidence, with cross-run comparisons treated as secondary.

Future work. Four directions follow: (1) model-adaptive evidence presentation to address the Opus regression on strong-baseline categories, (2) agentic graph traversal to improve the near-zero "change" category via targeted cross-admission queries, (3) expanding Tier C to 20+ patients for sufficient statistical power, and (4) two-physician blinded adjudication on a stratified hard subset.

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A DR.KNOWS Benchmark Results

We evaluate EpiKG on the DR.KNOWS diagnostic reasoning benchmark [30], which measures knowledge graph traversal accuracy using real PostgreSQL-backed clinical data.

Metric	Score	% of Baseline
Overall	0.420	49.7%
Path Discovery	44.4%	—
Reasoning Accuracy	22.2%	—
1-hop Traversal	50.0%	—
2-hop Traversal	25.0%	—
3-hop Traversal	0.0%	—

Table 10: DR.KNOWS benchmark results. Multi-hop accuracy degrades due to PostgreSQL CTE traversal limitations versus native graph databases—an intentional architectural tradeoff for ACID compliance and transactional guarantees.

The 3-hop traversal failure reflects the PostgreSQL CTE approach, which requires explicit recursive query construction rather than native graph pattern matching. This is an intentional design decision: EpiKG prioritizes ACID compliance and operational simplicity (single-database deployment) over raw graph traversal performance.

B MedQA-USMLE Baseline

To establish the baseline LLM capability, we evaluated the system's language model (Claude Opus 4.5) on MedQA-USMLE [25] without any retrieval augmentation.

Model	Accuracy	Step 1	N
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 11: MedQA-USMLE results. Our LLM-alone baseline is competitive, confirming that performance gains from KG augmentation reflect the system's contribution rather than a stronger base model.

C Scalability Metrics

Metric	Value
Documents processed	145
Patients	85
KG nodes	3,100
KG edges	8,803
Shared concept dedup ratio	∼36 nodes/patient
1-hop traversal latency	0.57 ms
2-hop traversal latency	0.75 ms

Table 12: Scalability metrics at current deployment scale. Sub-millisecond traversal latency enables real-time clinical decision support.

D System Component Scale

E Human Validation Pilot

To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-certified physician (specialty withheld for double-blind review) who scored system outputs on a 4-point scale (correct, partially correct, incorrect, clinically dangerous) without knowledge of condition labels.

Component	Scale
Clinical calculators	201
Guideline sections (RAG corpus)	1,202
OMOP vocabulary relationships	20M+
NLP ensemble models	3
Assertion trigger patterns	101
KG edge types	24
KG node types	13
Assertion classes	7

Table 13: System component scale. The OMOP vocabulary provides 20M+ concept relationships for multi-hop graph traversal.

Pilot findings ($n = 5$, single reviewer). The pilot covered 5 ClinicalBench C1 pairs and revealed:

- The automated LLM-as-judge scoring was overly strict on uncertainty and sequence questions, rating partially correct answers as incorrect.
- Gold standard answers were assessed as fully correct in 3/5 cases, partially correct in 1/5, and requiring revision in 1/5.
- Model answers were rated partially correct or better in 3/5 cases where the auto-scorer marked them incorrect.

Planned full adjudication (not yet complete). A two-physician blinded adjudication targeting $n \geq 30$ pairs stratified across conditions (C1/C4 for ClinicalBench; B2/B3 for Slice Bench) is planned, with predefined endpoints: (1) inter-rater κ, (2) human–LLM judge agreement rate, (3) discordance taxonomy (false strict, false lenient, genuine disagreement). This study is in preparation and results are not included in the current paper.

F Reproducibility Details

Models. ClinicalBench main ablation (Table 4): MedGemma 27B (answering) with LLM-as-judge scoring. ClinicalBench cross-model (Tables 7, 8): MedGemma 27B and Claude Opus 4.6 (claude-opus-4-20250514) with deterministic keyword evaluator. Slice Bench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929, answering), Claude Opus 4.6 (judging). All experiments used temperature 0 for reproducibility.

Evaluation. The cross-model results use a deterministic keyword evaluator that matches temporal and assertion keywords (e.g., "was"/"previously" for historical, "has"/"current" for present) against gold-standard categories. This evaluator strips echoed evidence preambles before scoring current_state/historical categories to prevent cross-contamination, and uses expanded uncertainty keywords including medical hedging terms ("likely", "probable", "concerning for").

Bootstrap protocol. All confidence intervals computed with $n = 2{,}000$ bootstrap resamples, seed 42, bias-corrected and accelerated (BCa) method [28].

Data. All experiments used de-identified MIMIC-IV clinical notes under the PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Computational resources. ClinicalBench: approximately 3.6 hours on a single machine with GPU (MedGemma inference via Ollama); Opus cross-model validation approximately 20 minutes via Anthropic API. Slice Bench: approximately 2 hours using Anthropic API.

G Assertion Category Definitions

H Slice Bench Tier-Stratified Results

I ClinicalBench Full Per-Category Results

J Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [26] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. All clinical questions were

Assertion	Definition	Example
PRESENT	Condition currently affirmed	“has diabetes”
ABSENT	Condition explicitly negated	“denies chest pain”
POSSIBLE	Condition suspected, not confirmed	“possible pneumonia”
CONDITIONAL	Contingent on specific circumstances	“if febrile, start antibiotics”
HYPOTHETICAL	Discussed as a scenario	“would need dialysis if. . . ”
FAMILY_HISTORY	Attributed to a family member	“mother had breast cancer”
HISTORICAL	Previously true, not necessarily current	“former smoker”

Table 14: The 7-value assertion taxonomy used in EpiKG, extending the i2b2 2010 6-class standard by separating HISTORICAL from FAMILY_HISTORY.

Table 15: Slice Bench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution. The KG benefit concentrates in Tier C.

		Score (%)					
Tier	Encounters	B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

Category	C1	C2	C3	C4	C5
Negation	100.0	93.6	81.8	100.0	98.2
Conditional	100.0	90.0	55.0	65.0	70.0
Uncertainty	55.0	2.5	20.0	62.5	52.5
Family History	0.0	0.0	53.3	70.0	56.7
Sequence	80.0	80.0	80.0	80.0	72.5
Current State	64.0	54.0	50.0	64.0	68.0
Duration	100.0	100.0	83.3	53.3	63.3
Historical	94.0	78.0	68.0	72.0	60.0
Change	0.0	0.0	3.3	3.3	3.3
Overall	**73.2**	**62.5**	**60.5**	**71.5**	**68.2**

Table 16: Complete ClinicalBench per-category accuracy (%, deterministic keyword evaluator) for all five conditions. C4 shows the largest gains on family history (+70 ppvs C1/C2) and uncertainty (+7.5 ppvs C1).

generated programmatically from structured data; no new clinical annotations were created. The LLM-as-judge evaluation protocol does not involve human subjects. The system is designed for clinical decision *support*, not autonomous clinical decision-making, and is not intended to replace physician judgment.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.2. All claims include confidence intervals and are scoped with "to our knowledge" qualifiers where appropriate.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **Yes.**

Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for Slice Bench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, category-level trade-offs (C4 vs C1), and the "change" category capability gap.

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4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: **Yes.**

   Justification: Appendix F reports exact model versions (MedGemma 27B, Claude Sonnet 4.5, Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2{,}000$, seed 42, BCa method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes question generation and evaluation protocols.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: **No.**

   Justification: MIMIC-IV data is available under PhysioNet credentialed access (not open). Code will be released upon acceptance. Benchmark question templates and evaluation scripts will be included in the release.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: **Yes.**

   Justification: Section 4 specifies benchmark design, question counts, condition definitions, patient selection criteria, and evaluation methodology. Appendix F provides model versions and computational requirements.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly?

   Answer: **Yes.**

   Justification: Table 6 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Figure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample sizes ($n = 2$ patients per tier). The overall B2→B3 CI crossing zero is explicitly noted.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: **Yes.**

Justification: Appendix F reports computational requirements: ClinicalBench ≈3.6 hours (single GPU for MedGemma inference), Slice Bench ≈2 hours (Anthropic API).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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   Justification: The study uses de-identified clinical data under an approved data use agreement (Appendix J). No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous decision-making.

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    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: **Yes.**

    Justification: Section 6 discusses the potential for improved clinical decision support (positive) and risks of over-reliance on automated systems, context dilution, and the need for physician oversight (negative). Appendix J addresses data ethics.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

    Answer: **Yes.**

    Justification: The system operates on de-identified data under PhysioNet DUA. It is designed as a decision support tool requiring physician oversight. Code release will include usage guidelines emphasizing the system is not validated for autonomous clinical use.

12. **Licenses for existing assets**

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    Answer: **Yes.**

    Justification: MIMIC-IV is cited [26] and used under PhysioNet Credentialed Health Data Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open standard.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **Yes.**

    Justification: ClinicalBench and Slice Bench are new benchmarks fully documented in Section 4, with question generation procedures, category definitions (Appendix G), and evaluation protocols. Templates and scoring scripts will accompany the code release.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

Answer: **N/A.**

Justification: No crowdsourcing was used. Physician adjudication (Appendix E) involves expert review by co-authors, not human subjects research.

15. **IRB approvals or equivalent**

Question: Does the paper describe potential risks and does it include the IRB approval number if applicable?

Answer: **N/A.**

Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not require separate IRB approval per PhysioNet policy.